

SOC Analyst Hands-On Project Report

Title: Enterprise Threat Detection & Investigation Using Splunk

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Role: SOC Analyst (Hands-On Project Simulation)

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1. Executive Summary

This project simulates a real-world **Security Operations Center (SOC)** environment where multiple attack activities were executed, logged, ingested into Splunk, and analyzed using **core SPL queries**.

As part of the investigation, the following threats were detected:

- Brute-force login attempts
- Successful unauthorized login
- Malware execution
- Suspicious PowerShell activities
- Lateral movement using PsExec & WMIC
- Persistence mechanisms (Scheduled tasks & Registry changes)
- Outbound communication to a malicious IP

This report documents the logs, SPL queries, findings, attack timeline, and remediation recommendations.

2. Project Objective

The objective of this project is to:

- Build a simulated SOC environment
 - Ingest Windows and Sysmon logs into Splunk
 - Use SPL queries to detect suspicious activities
 - Investigate each alert like a SOC Analyst L1
 - Create an incident timeline
 - Produce a professional SOC investigation report
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3. Tools & Log Sources

Tool / Source		Purpose
Splunk Enterprise		Log analysis, searches, threat detection
Windows Logs	Security	Login events, scheduled tasks, registry changes
Sysmon Logs		Process creation, command execution, network connections
Notepad		Log creation
CSV/Excel		Timeline creation

4. Attack Scenarios Simulated

Scenario	Description
Brute-force Attack	Multiple failed login attempts on admin account
Unauthorized Login	Successful login after brute-force

Malware Execution	Update.exe executed from unknown source
Suspicious PowerShell	Encoded commands detected
Lateral Movement	Psexec used to access remote system; WMIC enumeration
Persistence Mechanisms	Scheduled tasks & registry run key added
C2 Communication	Outbound network connection to suspicious IP

5. SPL Queries Used in Investigation

Below is a subset of the 15 essential SPL queries used:

Failed Login Attempts

```
index=windows_security EventCode=4625  
| stats count by AccountName, SourceIp  
| where count>3
```

Successful Login

```
index=windows_security EventCode=4624
```

Malware Execution

```
index=sysmon_process process_name="Update.exe"
```

Suspicious PowerShell

```
index=sysmon_process process_name="powershell.exe"
```

Psexec Remote Execution

```
index=sysmon_process process_name="psexec.exe"
```

WMIC Command

index=sysmon_process process_name="wmic.exe"

Scheduled Task Creation

index=windows_security EventCode=4698

Registry Persistence

index=windows_security EventCode=4657 RegistryKeyPath="*Run*"

Suspicious Network Connection

index=sysmon_network EventCode=3

6. Incident Timeline

Time	Event	User	Target	Details
01:00	Failed Login	admin	10.1.10.100	Multiple failed attempts detected
01:05	Successful Login	admin	10.1.10.100	Possible brute-force success
02:15	Malware Execution	svc-back up	Update.exe	Malicious file executed
02:16	Suspicious PowerShell	svc-back up	powershell.exe	Encoded command executed
02:30	PsExec Movement	Lateral svc-back up	10.1.10.22	Remote execution detected
02:32	WMIC Enumeration	Remote svc-back up	10.1.10.22	System info retrieved
02:33	Unauthorized Login	svc-back up	WIN-FILE01	Lateral access

03:00	Scheduled Created	Task	svc-backup	Update.exe	Persistence mechanism
03:05	Registry Modified	Run Key	svc-backup	HKCU...\Run\Update	Startup persistence
03:15	Outbound C2 Traffic		svc-backup	45.82.19.11	Possible exfiltration

7. Findings

- ✓ Admin account was brute-forced and compromised
- ✓ Malware (Update.exe) executed by service account
- ✓ Lateral movement performed using PsExec and WMIC
- ✓ Attacker added persistence using scheduled tasks & registry run key
- ✓ Outbound traffic to suspicious external IP (C2 activity)

Attack chain observed matches the MITRE ATT&CK tactics:

- Initial Access
 - Execution
 - Persistence
 - Privilege Escalation
 - Lateral Movement
 - Command & Control
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8. Recommendations

- ♦ Block the suspicious IP (45.82.19.11)
- ♦ Reset passwords for admin and svc-backup accounts
- ♦ Enable MFA for all admin accounts
- ♦ Remove unauthorized scheduled tasks and registry keys

- ◆ **Restrict PsExec and WMIC usage**
 - ◆ **Implement endpoint logging & antivirus scanning**
 - ◆ **Monitor PowerShell encoded command usage**
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9. Skills Demonstrated

- Log ingestion and normalization
- Writing and executing Splunk SPL queries
- Windows and Sysmon log analysis
- Threat detection and correlation
- Lateral movement investigation
- Building an incident timeline
- Writing a SOC incident report

This project demonstrates readiness for:

- ✓ SOC Analyst L1
- ✓ Cybersecurity Analyst
- ✓ Security Monitoring Analyst
- ✓ SIEM Operator